

Tesneem Essa

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EDUCATION

Macaulay Honors at Hunter College

New York, NY

Bachelor of Arts in Computer Science | GPA: 3.94/4.00

Jun 2025

Relevant Coursework: Software Analysis & Design, Discrete Math, Computer Architecture, Operating Systems, Web Development, iOS Development, Digital Product Development, Computer Theory, Physics I, Calculus II, Matrix Algebra, Capstone, VR, AR, & MR

SKILLS

Programming: C++, Python, Pandas, SQL, HTML, CSS, JavaScript, SwiftUI, React, TensorFlow, REST API, Dart, C#

Technologies: Xcode, VSCode, Git/GitHub, Excel, Linux, Hugging Face, Firebase, Unity

Spoken Languages: Arabic (Conversational)

PROJECTS

PlayPaws ([Link](#))

May 2025

- In a team of four, developed a puppy playdate app using Flutter and Python.
- Created a dog-matching algorithm using sentence-transformer embeddings, integrated with a backend API, and implemented real-time playdate scheduling with Firestore.

In the Dark ([Link](#))

May 2025

- Using Unity, developed a VR horror game scene with a focus on interactivity and immersion.

AI vs Real Images ([Link](#))

Dec 2024

- Collaborated with a team of three to develop machine learning models for detecting AI-generated versus real images.
- Utilizing Python and TensorFlow Keras, implemented EfficientNet-B3 and B0 architectures with EarlyStopping, and ReduceLROnPlateau callbacks to prevent overfitting and optimize training.
- Achieved 98% accuracy on unseen data and deployed an interactive web app using Streamlit on Hugging Face.

Netflix Recommendation System ([Link](#))

Nov 2024

- Integrated Blip Image Model, Sentence Transformers, and embedded documents to deliver tailored recommendations based on title, description, and genre of Netflix shows and movies.
- Developed a Retrieval-Augmented Generation (RAG) system in Python for fast and efficient recommendations.
- Awarded Best Project in Class.

RELEVANT EXPERIENCE

Cambio Labs

New York, NY

AI & Data Science Intern

Jun 2025 - Aug 2025

- Built GPT-based Skill Assessment Assistant analyzing 500+ responses into skill scores and class insights.
- Created scripts and dashboards (radar charts, summaries) for educators to track student growth.
- Automated grant data extraction with regex chunker, structuring 1,000+ records from 60+ documents.
- Developed RAG chatbot with MongoDB & Hugging Face to draft context-rich grant responses.

CUNY Tech Prep

New York, NY

Data Science Fellow

Jun 2024 - May 2025

- Selected for a competitive data science fellowship with students from across the 11 CUNY senior colleges, where Fellows create technical projects using tools such as Python 3, Jupyter Notebooks, Pandas, Numpy, Scikit-learn, and SQL.
- Participate in weekly courses and learn industry best practices for exploratory data analysis (EDA), feature engineering, data collection and processing, statistical modeling, data visualization, machine learning techniques, data science process, and big data.

LinkedIn

New York, NY

Software Engineer Intern

Jan 2025

- Collaborated with a team of five to develop a video-clipping app in three weeks in response to a user requested feature for the LI Events Team.
- Utilizing ffmpeg.wasm and Next.js, created three editing modes: 30-sec clips, slider-based trimming, and time-stamp cutting.
- As the infrastructure engineer, integrated dependencies, set up ffmpeg.wasm, implemented the clipping and slider features, and enabled video playback within the selected range.